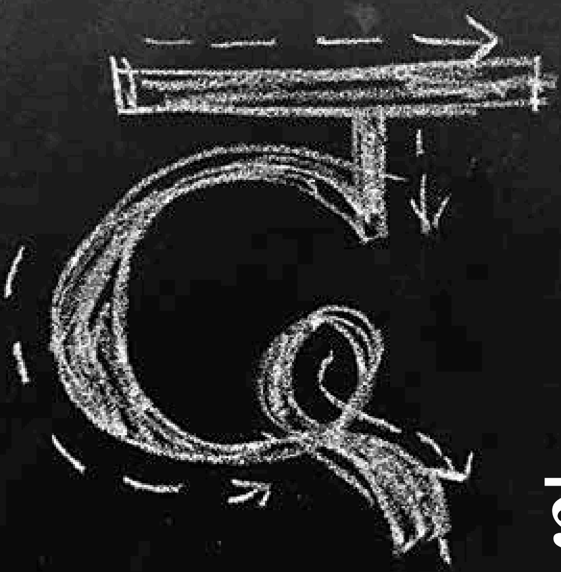
CDAC keyboard
The .apk file for the CDAC keyboard can be downloaded directly from their website.
https://dgit.in/sep19-39

Inscript layout
Download the INSCRIPT keyboard layout and convert your QWERTY keyboard to an Indic keyboard **https://dgit.in/sep19-40**



designing indic keypads for smartphones

There are many challenges with just indic text input

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The scripts for most of the Indian languages originated from a script called brahmi. The edicts of Ashoka used this script in the 3rd century BC. It has evolved into the brahmic family of scripts, which is used to write many languages in the subcontinent, Southeast Asia and even Japan. The important feature about the brahmic family of scripts is that in it, there is a single unit representing a sequence of consonants and vowels. Two consonants may also be joined together to form conjuncts. By contrast, in the latin alphabet, the vowels are given their own separate letters, and are as important as the consonants. The local languages also tend to have large character sets - Malayalam for example, has more than sixty characters, all of which are quite difficult to fit onto the limited real estate on smartphone screens.

CONSIDERATIONS AND CONSTRAINTS

Vivekanand Pani, co founder and CTO of Reverie Language Technologies, explains the considerations for designing a keypad for Indic languages, “the English alphabet has only 26 letters, while Indian languages typically have a lot more. There are twelve to fourteen vowels, and there are at least 35 consonants, sorted into vargas of five each. There are as many matras as vowels. And then halant, nukta, those characters are there. That makes the alphabet set pretty large. It becomes difficult to accommodate so many characters within a small space. Because the alphabet set is large, it appears intimidating.”

Juggling the available real estate can be quite a hassle, especially if the designers of the keyboards start off with the QWERTY layout. Pani explains, “in the English keypads, the letter layout is more or less the same as the QWERTY keyboard from typewriters. In mobile phones what has happened is that a lot of other keys have been moved into different layers, so that in the basic English layout, you see only letters and most commonly-used punctuations. That way, what happens is that you use a lot fewer keys, and therefore you are able to give more real estate to each key. On the top row, you have only ten letters. If a row has only ten letters, then each row potentially can have one tenth of the screen width space. Similarly, it has got only three rows. When it has three rows, you have the rest of your screen available to look at the application, or what you are typing. Your keyboard should not take a lot of area, and it is still having enough space for each key so that somebody is able to type without making many errors.”

When you break down the process, writing in Indic languages is not as simple as stringing a number of alphabets together. Pani says, “when you are designing a keypad, one thing that